CROSS-ASSET QUANT RESEARCH

Peso vs. Copper

USDCLP vs. Copper

Trade: + CLP / **-** COPPER | **Universe:** FX ↔ Commodity | **Horizon:** 3–6M | **Conviction:** MODERATE

- ***Copper*** *is Chile's biggest export and sits near a* ***record***
- *Yet the* ***peso is down 8%*** *in six months while copper is* ***up 18%***
- *A* ***global dollar and US-yield bid****, not Chile's trade, is doing the damage*

SUMMARY	
Current Gap	**−1.4σ**
Beta	**+0.15**
Lookback	**~23 yrs**
Times This Wide	**7% of days**
Long Leg	**Chilean peso**
Short Leg	**Copper**
Target	0.0σ
Stop	−2.5σ
R/R	**1.3x**

Data source: Yahoo Finance daily closes (CLP=X, HG=F), Dec 2003–2 Oct 2026
Intuition: Copper → Chile's export income → peso (commodity-currency link); Rio Times, 2 Oct 2026

Executive Summary

RECOMMENDATION: LONG CLP / SHORT COPPER | Gap: −1.4σ | Target: 0.0σ | Stop: −2.5σ | Horizon: 3–6M | MODERATE CONVICTION

01

Economic Mechanism

- **Copper** is Chile's biggest export; Chile mines nearly **a quarter** of world supply
- More copper income means more **export dollars**, so the peso should **firm**: a classic **copper currency**
- **Comex copper** is the real-time read on that income
- **The leak:** Chile imports oil, and a **US-yield or dollar spike** hits the peso whatever copper does

02

Quantified Dislocation

- Over 6 months copper **+18%**, peso **−8%**
- At the usual link (beta **0.15**) the peso would be **~9.5% stronger**
- Gap **−1.4σ**, seen on about **1 day in 13**
- Only **−0.8σ** on 1 Oct; on **12–36 months** just −0.6σ to −0.9σ
- A **recent** dislocation, not a long-running one

03

Trade & Failure Conditions

- **Long peso / short copper futures**, beta-hedged (1 : 0.15)
- **Target:** gap back to average, **+9.5%** (USDCLP ~903)
- **Stop:** −2.5σ, about **−7.0%**; risk/reward **~1.3x**, so size modestly
- **Fails if** US yields keep rising and EM currencies keep falling
- **Or if** Chile's mining output keeps shrinking

Relationship & Economic Rationale

01 Economic Mechanism

- **Copper price up**: miners (Codelco, Escondida, Antofagasta) book more **dollar revenue**
- They **sell dollars for pesos** (wages, taxes, royalties) and foreign money follows
- **Peso demand rises**, USDCLP falls; **Comex copper** is the real-time read on that income

02 Transmission Channels

- **Trade balance (+):** copper is the largest export; ~¼ of world supply
- **Capital flows (±):** copper inflows (+); US yields and the dollar pull money out of EM (−)
- **Policy (+):** BCCh 4.50% vs Fed 3.75–4.00%: positive **carry**

03 Why the Link Should Persist

- Export income **must be converted** into pesos (wages, taxes, royalties)
- A dollar or US-yield spike can **hide** this for weeks, not permanently
- **Tonnage matters:** July output **−9%** (storms), so the windfall is smaller than the price implies

Key Numbers

Asset Y	**Chilean peso**
Asset X	**Copper (HG)**
Data Range	**2003–2026**
Data Source	**Yahoo Finance**
Measure Used	**Beta-adj. gap**
Correlation (Full)	**+0.32**
Correlation (1Y)	**+0.32**
Avg Gap	**−1.6%**
Current Gap	**−10.7%**
Current Z-Score	**−1.4σ**
Largest Past Gap	**−5.0σ**

Historical Evidence

How We Measured It

- **Gap** = peso 6-month return − **beta × copper** 6-month return, as a **z-score**
- **Beta 0.15** = slope of weekly peso returns on weekly Comex copper returns
- **Data:** Yahoo Finance daily closes, Dec 2003–Oct 2026; peso in USD per CLP (up = stronger)
- **6 months** because the trade runs 3–6 months; other windows on slide 5

The Chart



Does the Relationship Hold Over Time?

Period	Corr	Avg Gap	Max Gap	#>2σ	Held?
Full Sample	+0.32	−1.6%	5.0σ	11	Y
Last 10Y	+0.39	−2.4%	3.5σ	7	Y
Last 5Y	+0.40	−2.7%	3.5σ	5	Y
Last 3Y	+0.41	−3.1%	2.3σ	2	Y
Last 1Y	+0.32	−0.3%	1.8σ	0	Y
GFC 2008–09	+0.25	−1.6%	5.0σ	3	N
Overall	**+0.35**	**−1.6%**	**5.0σ**	**11**	**5 of 6**

- ***Stable, with a leak:*** *weekly correlation ≥ 0.30 in **5 of 6** windows (0.32 full, 0.64 in 2024)*
- ***Weak spot:*** *only 0.25 in the 2008–09 crash*
- ***Can widen first:*** *−5.0σ (Nov 2008), −2.9σ (Jul 2022, when the BCCh stepped in)*

Current Dislocation — Why Now?

WHY IS THE RELATIONSHIP DISLOCATED NOW?

Size of the Deviation

- **−1.4σ** on 6m: only **7%** of days were this wide
- At the average gap the peso is **~9.5% stronger** (USDCLP ~903)
- **1m −1.8σ, 3m −1.5σ**, but 12m −0.6σ, 24m −0.9σ, 36m −0.6σ
- Was **−0.8σ** on 1 Oct

What Caused It

- **The peso, not copper:** peso −5.5%, copper +1.3% in a month
- **Fed** hiked to 3.75–4.00% (16 Sep); long US yields at **highest since 2002**
- **Mexico and Brazil** fell too: mostly a **global dollar** move
- Chile's soft data (July output **−9%**) adds to it

Which Market Is Right?

- **Copper is right on price:** near records, smelter outages, low stocks
- **Peso is right on volume:** July output lowest for the month since 2011
- But **Comex +36%** in 12m vs tonnage −9%: income far higher, peso −3%

Asymmetric Setup

- **+9.5%** if the gap closes vs **−7.0%** at a −2.5σ stop (**~1.3x**)
- On the **3-year average gap:** +7.9% (~1.0x), so modest
- A −2.0σ stop shows 2.3x but is only **0.6σ away** (one day's move)
- **Carry** about +0.15% a quarter

NORMALIZATION CATALYSTS

Oct 27–28: BCCh and FOMC — 0 – 1 m

- **Oct 27 BCCh:** 4.50% since Dec 2025; 'meeting by meeting' after CPI +0.6%
- **Oct 28 FOMC:** hold expected; **December hike** per the dots

Expected: gap narrows to c.−1.0σ

Dollar and yield peak; Chile data — 1 – 3 m

- **US long yields and dollar** (DXY ~101) stop rising; MXN, BRL, CLP rebound together
- **Data:** Sept copper output (end Oct), Imacec (early Nov)
- **Dec 9 FOMC**, **Dec 15 BCCh**

Expected: gap narrows to c.−0.5σ

Mine recovery; tariff clarity — 3 – 6 m

- **Mines restart:** Los Pelambres and Caserones recover after the storms
- **Export values:** BCCh monthly copper export value turns positive again as output recovers

Expected: gap narrows to c.0.0σ

Market vs. Normal Relationship

What the Market Says vs. What the Relationship Says

Metric	Today	If Normal	Gap	Why It Should Close
USDCLP (pesos per $)	990	903	−8.7%	• **903** = USDCLP if the 6m gap were at its average • Beta **0.15**, copper held flat
Ratio / Spread	−1.4σ	0.0σ	+1.4σ	• **1.4σ** below the 6-month average gap
Macro Story in Copper	Scarce	—	—	• Comex **+36%** in 12m, **+77%** in 3 yrs; near **records** • Chile's own July output **−9%** (storms)
Macro Story in Peso	Weak	—	—	• Peso **−8%** in 6m, a **13-month low** (USDCLP ~990) • **Dollar and US yields**, not copper: the inconsistency
Correlation	+0.32	+0.32	0.00	• Last 1Y vs full sample: **link intact** (0.64 in 2024, 0.30 in 2025, 0.38 in 2026)

Source of Edge

Why the Gap Exists

- **Global dollar and US-yield** move hits all EM currencies at once
- Copper is priced on **physical scarcity** (outages, low stocks)
- Peso also trades on Chile's **weak output** and rate path; USDCLP **+5.5%** in a month

Why It Hasn't Closed

- **Different investors** trade each leg: macro funds sell EM on US yields, metals traders buy copper on supply
- **Weak Chile data** (August activity −1%, unemployment 9.6%) gives dollar buyers a reason to keep going

What Proves You Right

- **USDCLP toward ~903** as US yields stop rising and EM FX recovers, copper near $6.5/lb
- **First tests:** Oct 27 (BCCh), Oct 28 (FOMC)
- **Wrong if** yields keep rising and the peso falls with them

Trade Expression

POSITION PARAMETERS

Structure	**LONG CLP / SHORT COPPER**
Instruments	**USDCLP NDF / Copper futures**
Sizing	**Beta-hedged: 1 : 0.15**
Entry	**−1.4σ (USDCLP 990)**
Target	**0.0σ (+9.5%)**
Risk/Reward	**1.3x**
Stop	**−2.5σ (−7.0%)**
Horizon	**3–6 months**
Exit Plan	**Take profit at 0.0σ**

TRADE RATIONALE

Why This Expression

- **Long peso alone** is a bet on the dollar and EM risk
- **Short copper** (beta 0.15) strips the commodity move; **US-yield risk stays**

What Convergence Looks Like

- **Mostly the peso catching up:** USDCLP 990 → ~903 if copper is flat
- **Copper falling back** also closes the gap and pays on the short leg

Cost of Holding

- **Peso leg ~+0.6% a year** (BCCh 4.50% vs Fed 3.75–4.00%), ~+0.15% a quarter
- **Copper roll** small

Liquidity

- **USDCLP NDF** (deep offshore) or onshore spot; **Comex HG or LME** copper futures
- **LME** tracks Chile's realised price better than Comex

Thesis Invalidation

- **Gap** hits −2.5σ, or **52-week correlation** below 0.10 for 8 weeks
- **US long yields** make a new high after the Oct 28 FOMC

Failure Conditions — When Are We Wrong?

Risk	Sev	Prob	Description	Mitigant / Hedge
Structural Break	MED	LOW	• **Link failed** in 2005–06 (0.07) and 2009 (−0.15); 0.30 in 2025 • **Falling output** (July −9%) weakens the income channel: higher copper no longer means more export dollars	• Track the **52-week correlation**; exit if below **0.10 for 8 weeks** • Watch monthly **INE copper volumes**
Regime Shift	HIGH	MED	• **US-yield regime:** Fed hiked to 3.75–4.00% (16 Sep), dots point to one more; long yields highest since 2002 • EM currencies then fall whatever commodities do (**2022: −2.9σ**)	• Short copper offsets only a little (**beta 0.15**) • **Trim size** into the FOMC (Oct 28, Dec 9); keep it small
Hidden Driver	HIGH	HIGH	• **US yields and dollar**, not copper, set the peso today (Mexico, Brazil fell too) • **Comex tariff premium** collapsed 10 Sep (spread −$20/t): Comex can move without Chile's price	• Watch **DXY, 10-year yield, USDMXN** as third signals • **Hedge with LME** if the spread keeps moving • If yields keep rising, the gap is **fair value**, not a dislocation
Policy Intervention	MED	LOW	• **BCCh supported the peso** before (Jul 2022, US$25bn): helps us • **Risks:** a dovish surprise (unlikely, inflation 4.1%) or a **US copper tariff** decision repricing Comex	• **Small size** into Oct 27 (BCCh) and any US tariff news • **Prefer LME** copper for the hedge
Divergence Widens	HIGH	MED	• Gap hit **−2.9σ** (Jul 2022) and **−5.0σ** (Nov 2008) before closing • Our −2.5σ stop is only **1.1σ** away	• **Stop −2.5σ** and a **6-month time limit** • **Build in thirds**; do not add as it widens • **Carry +0.15%** a quarter pays a little while we wait